Document register and engineering-query register

Flash drum D-400 unit · every document of the unit, the dependencies between them, the engineering queries and the cross-document checks

Document no. FD-V1-PM-DOC-001

The master list of the documents of the D-400 unit with their number, discipline, revision, status and file; the deliverable tree (which document is built from which) and the design inputs each document uses; the register of the engineering queries raised while the documents were generated, with the later documents that answer them; and the cross-document checks computed by the Project & deliverables tab.

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Holds and engineering queries

- EJ-01 Two numbering schemes and two status words
- EJ-02 The SRS is not a controlled file
- EJ-03 Consequence screening used line sizes assumed before the line list
- EJ-04 P&ID Rev 1 collects 18 open queries
- EJ-05 Isometrics drawn at sizes that an open query changes
- EJ-06 Two fire relief loads for PSV-400
- EJ-07 D-400 can drain back through FCV-400 after a trip
- EJ-08 Fire case assumes drainage that the layout does not show

Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

Rev	Date	Description	Originator	Checked	Approved
0	28-OCT-26	Issued for teaching	A. Soottitantawat	—	—

DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-PM-DOC-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Basis

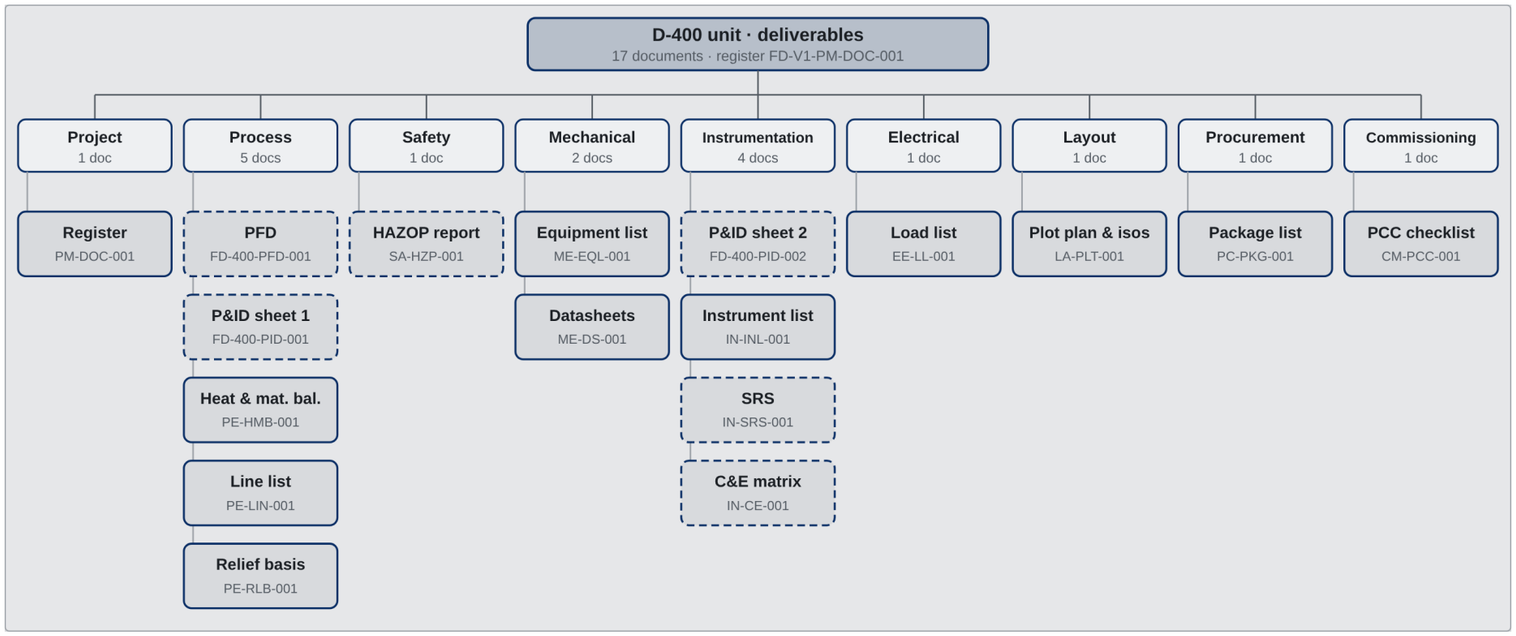
Numbering	FD-V1-(discipline)-(type)-001: PM project, PE process, SA safety, ME mechanical, IN instrumentation, EE electrical, LA layout, PC procurement, CM commissioning. The PFD and the P&ID sheets keep their earlier numbers FD-400-PFD-001, FD-400-PID-001 / -002 (EJ-01).
Revision and status	Every document is Rev 0 · Issued for teaching; none has been checked or approved (CHK and APP blank). Letters A, B … would mark issues before approval (IFR, IFA), numbers from IFC.
Roles	Originator prepares and raises queries; checker (same discipline) checks; approver (lead engineer) approves; document control numbers, issues, transmits and keeps this register; the owner reviews each issue (code 1 no comment, 2 proceed with comments, 3 revise and resubmit, 4 information).
Queries	Raised by the discipline that finds the problem; answered by the owner of the input; closed when the documents the answer changes are re-issued. Statuses: OPEN, PARTLY ANSWERED, ANSWERED (a later document gives the answer; the query stays until the revision is issued).
Source	Register and queries from epc/epc.json (generators epc/mk_epc.js, mk_epcB.js, mk_epcP.js) and the Layout tab; page counts read from the files; checks computed by the Project & deliverables tab of the simulator page.

2 · Document register

Document no.	Title	Discipline	Rev	Status	Pages	File	Taught in (tab)	Typical phase · status path	Open queries
FD-V1-PM-DOC-001	Document register and engineering-query register	Project	0	Issued for teaching	8	FD-V1-PM-DOC-001.pdf	Project & deliverables	all phases · monthly issue · final register at handover	EJ-01
FD-V1-PE-HMB-001	Heat and material balance (design, maximum rate, turndown)	Process	0	Issued for teaching	7	FD-V1-PE-HMB-001.pdf	Process engineering	FEED · IFR → IFA · IFC	EP-02, EP-04, EP-08
FD-V1-PE-LIN-001	Line list and line sizing	Process	0	Issued for teaching	5	FD-V1-PE-LIN-001.pdf	Process engineering	FEED (draft) · IFA · IFC	EP-01, EP-03, EP-05, EP-07, EP-11, LA-03, LA-06
FD-V1-PE-RLB-001	Relief load basis and PSV-400 sizing	Process	0	Issued for teaching	6	FD-V1-PE-RLB-001.pdf	Relief & vent	FEED · IFA · IFC	ER-01, ER-02, ER-03, ER-04, ER-05, ER-06, ER-07, EJ-08
FD-V1-ME-EQL-001	Equipment list	Mechanical	0	Issued for teaching	3	FD-V1-ME-EQL-001.pdf	Mechanical	FEED · IFA · IFC	—
FD-V1-ME-DS-001	Equipment datasheets (D-400, E-400) with GA sketches	Mechanical	0	Issued for teaching	5	FD-V1-ME-DS-001.pdf	Mechanical	FEED · IFP (enquiry) · certified vendor data	EM-02, EM-03, EM-04, EM-05, LA-02
FD-V1-IN-INL-001	Instrument list and I/O count	Instrumentation	0	Issued for teaching	8	FD-V1-IN-INL-001.pdf	Instrumentation & control	Detailed engineering · IFA · IFC, as-built	EP-09, EI-01, EI-02, EI-03, EI-04, EI-05, EI-06
FD-V1-EE-LL-001	Electrical load list	Electrical	0	Issued for teaching	6	FD-V1-EE-LL-001.pdf	Electrical	Detailed engineering · IFA · IFC	EE-01, EE-02, EE-03, EE-04, EE-05
FD-V1-PC-PKG-001	Procurement package list	Procurement	0	Issued for teaching	6	FD-V1-PC-PKG-001.pdf	Procurement to commissioning	Procurement · IFP · updated per order	EC-01
FD-V1-CM-PCC-001	Pre-commissioning and commissioning checklist	Commissioning	0	Issued for teaching	5	FD-V1-CM-PCC-001.pdf	Procurement to commissioning	Construction → commissioning · IFA · signed at MC / handover	EC-02, EC-03, EC-04
FD-V1-LA-PLT-001	Plot plan, elevations, access and piping isometrics	Layout	0	Issued for teaching	17	FD-V1-LA-PLT-001.pdf	Layout & isometrics	FEED → detailed · IFA (3-D model review) · IFC per isometric, as-built	EJ-05, EJ-08
FD-400-PFD-001	Process flow diagram	Process	0	Issued for teaching (data: Issued for training, EJ-01)	—	FD-400-PFD-001.pdf (published with the page)	PFD	Concept / FEED · IFR → IFA · IFC	—
FD-400-PID-001	P&ID sheet 1, D-400 and E-400	Process	0	Issued for teaching (data: Issued for training, EJ-01)	—	A3 drawing image in the P&ID tab (no PDF)	P&ID	FEED · IFR → IFA (HAZOP) · IFC, as-built	EP-06, LA-04, LA-07, EJ-04, EJ-07
FD-400-PID-002	P&ID sheet 2, control configuration, loop data, SIS C&E	Instrumentation	0	Issued for teaching (data: Issued for training, EJ-01)	—	A3 drawing image in the P&ID tab (no PDF)	P&ID	FEED · IFR → IFA · IFC, as-built	—
FD-V1-IN-CE-001	Cause and effect matrix	Instrumentation	0	Issued for teaching	—	FD-V1-IN-CE-001.pdf (published with the page)	C&E matrix	Detailed engineering · IFA · IFC	EJ-07
FD-V1-IN-SRS-001	Safety requirements specification	Instrumentation	0	Issued for teaching	—	in the SRS tab only (no PDF, EJ-02)	SRS of SIF	Detailed engineering · IFA · IFC (frozen before the SIS FAT)	EJ-02
FD-V1-SA-HZP-001	HAZOP report	Safety	0	Issued for teaching	—	FD-V1-SA-HZP-001.pdf (published with the page)	HAZOP	FEED · report on the IFA P&ID · close-out before IFC	—

17 documents: 11 generated by the Engineering tabs, 6 issued before them. 11 PDFs in the tab folders with 76 pages (this register counted at its previous build).

3 · Deliverable tree and dependencies



From	To	What flows
FD-V1-PE-HMB-001	FD-400-PFD-001	flags of streams 2–5 (design case), duty in the basis panel
FD-400-PFD-001	FD-400-PID-001	equipment, main lines, stream numbers
FD-V1-PE-HMB-001	FD-V1-PE-LIN-001	flow, density, viscosity, T and P of every line, three cases
FD-V1-PE-HMB-001	FD-V1-ME-EQL-001	duty, operating P/T, control-valve flows
FD-V1-PE-HMB-001	FD-V1-ME-DS-001	E-400 duty and LMTD, drum vapour / liquid loads
FD-V1-PE-HMB-001	FD-V1-PE-RLB-001	design feed, drum-liquid composition, blocked-outlet flows
FD-V1-PE-HMB-001	FD-V1-IN-INL-001	normal values, ranges, valve sizing flows
FD-V1-PE-HMB-001	FD-V1-EE-LL-001	P-400 absorbed power
FD-V1-PE-HMB-001	FD-V1-CM-PCC-001	performance-test guarantees
FD-400-PID-001	FD-V1-PE-LIN-001	line numbers, sizes as drawn, spec breaks

From	To	What flows
FD-400-PID-001	FD-V1-ME-EQL-001	equipment and valve tags
FD-400-PID-001	FD-V1-SA-HZP-001	nodes and design intent
FD-400-PID-001	FD-V1-IN-INL-001	instrument bubbles of sheet 1
FD-400-PID-001	FD-V1-LA-PLT-001	what is piped and where the in-line items sit
FD-400-PID-002	FD-V1-IN-INL-001	loops, alarms, cascades
FD-400-PID-002	FD-V1-IN-CE-001	SIS logic I-1 / I-2
FD-V1-SA-HZP-001	FD-V1-IN-SRS-001	deviations and safeguards (via LOPA) → SIF demands
FD-V1-IN-SRS-001	FD-V1-IN-CE-001	trip settings, delays, latches
FD-V1-IN-CE-001	FD-V1-IN-INL-001	trip rows, SIS I/O
FD-V1-IN-SRS-001	FD-V1-CM-PCC-001	proof tests, FAT / SAT of every SIF
FD-V1-ME-EQL-001	FD-V1-ME-DS-001	design P/T, materials
FD-V1-ME-EQL-001	FD-V1-PE-RLB-001	design pressure = PSV set
FD-V1-ME-EQL-001	FD-V1-PC-PKG-001	items to buy
FD-V1-ME-EQL-001	FD-V1-LA-PLT-001	sizes and weights
FD-V1-ME-DS-001	FD-V1-PE-RLB-001	wetted area, drum elevation
FD-V1-ME-DS-001	FD-V1-LA-PLT-001	nozzles and elevations (GA)
FD-V1-ME-DS-001	FD-V1-PC-PKG-001	enquiry data sheets
FD-V1-PE-RLB-001	FD-V1-PE-LIN-001	PSV inlet and tail-pipe lines
FD-V1-PE-RLB-001	FD-V1-PC-PKG-001	PSV-400 orifice
FD-V1-PE-RLB-001	FD-V1-LA-PLT-001	tail-pipe length and fall
FD-V1-PE-LIN-001	FD-V1-LA-PLT-001	line sizes, classes, design P/T for the isometrics
FD-V1-PE-LIN-001	FD-V1-PC-PKG-001	piping bulk
FD-V1-PE-LIN-001	FD-V1-CM-PCC-001	test pressures, test packs
FD-V1-IN-INL-001	FD-V1-EE-LL-001	DCS / SIS cabinet loads from the I/O count
FD-V1-IN-INL-001	FD-V1-PC-PKG-001	instrument and valve packages
FD-V1-IN-INL-001	FD-V1-CM-PCC-001	loop checks, calibration
FD-V1-EE-LL-001	FD-V1-PC-PKG-001	electrical package
FD-V1-EE-LL-001	FD-V1-CM-PCC-001	energisation checks
FD-V1-LA-PLT-001	FD-V1-PC-PKG-001	isometric bill of materials
FD-V1-LA-PLT-001	FD-V1-CM-PCC-001	test-pack boundaries per isometric
FD-V1-PC-PKG-001	FD-V1-CM-PCC-001	FAT and witness points

4 · Design inputs used by each document (ripple map)

R = a value printed in the document follows the input (revise when it changes); C = the document only checks against it. Documents built from a revised document (section 3) are re-checked. The tab recomputes the change for any new value of the input.

Document	Design feed rate (now 16 m³/h (K.Fd))	D-400 design pressure (now 450 kPa g (= PSV-400 set, P&ID "MAWP"))	PSV-400 set pressure (design pressure unchanged) (now 450 kPa g (K.Ppsv))	Hot-feed temperature (TIC-401 set point) (now 74 °C (K.TICsp))	Hot-oil supply temperature (now 180 °C (K.Tho))
FD-V1-PM-DOC-001 Document register and engineering-query register					
FD-V1-PE-HMB-001 Heat and material balance (design, maximum rate, turndown)	R — every stream, duty, valve openings			R — flash split, product compositions, duty	R — hot-oil streams H1–H5, return temperature
FD-V1-PE-LIN-001 Line list and line sizing	R — velocity and ΔP of every line	R — drum-side lines design P, class rating, test pressure		R — hot-feed and two-phase line densities	R — hot-oil lines design temperature (= Tho + 25)
FD-V1-PE-RLB-001 Relief load basis and PSV-400 sizing	C — blocked-outlet loads C1–C3 scale with the feed	R — set pressure = design → relieving pressure, loads, orifice	R — relieving pressure, loads, orifice	R — bubble pressure of the hot feed against the relieving pressure (C1–C3)	R — thermal relief of the blocked tube side (C7)
FD-V1-ME-EQL-001 Equipment list	R — E-400 duty, control-valve flows and Cv	R — D-400 design pressure, drum-side valves	R — PSV-400 row	R — E-400 duty	R — E-400 shell and XV-409 design temperature
FD-V1-ME-DS-001 Equipment datasheets (D-400, E-400) with GA sketches	R — E-400 rating (area excess), nozzle velocities	R — shell / head thickness, MAWP stamp, hydrotest	C — stamped MAWP ≥ set (UG-134)	R — E-400 rating	R — E-400 design temperature and rating
FD-V1-IN-INL-001 Instrument list and I/O count	R — FT-400 range and β, normal values, FIC-400 HI alarm	R — PT-420 range against the set (EI-01), PSV-400 row	R — PSV-400 row; PT-420 range against the set	R — TIC-401 normal value and alarms 69 / 79 °C	
FD-V1-EE-LL-001 Electrical load list	R — P-400 absorbed power				C — hottest surface against the T-class (T3, 200 °C) and the lowest AIT (225 °C)
FD-V1-PC-PKG-001 Procurement package list	R — FCV-400 / LCV-400 trims (PKG-03), E-400 (PKG-02)	R — D-400 (PKG-01) and PSV-400 (PKG-05) requisitions	R — PSV-400 requisition		C — E-400 and XV-409 ratings
FD-V1-CM-PCC-001 Pre-commissioning and commissioning checklist	R — performance-test guarantees	R — D-400 hydrotest, unit leak test (0.9 × set), PSV bench test	R — leak test 0.9 × set, PSV bench test	R — performance test TI-401 and product specs	
FD-V1-LA-PLT-001 Plot plan, elevations, access and piping isometrics	C — bottoms-line flashing check at FT-410, line sizes on the isometrics	R — design P on the drum-side isometrics			
FD-400-PFD-001 Process flow diagram	R — stream flags 2–5 (kg/h)			R — stream 3 flag 74 °C, E-400 duty in the basis panel	R — hot-oil flag 180 °C
FD-400-PID-001 P&ID sheet 1, D-400 and E-400	R — general note 1 (flows, duty) and note 13 (FIC-400 alarms)	R — "MAWP 450 kPa g" note, PSV-400 set	R — PSV-400 set on sheet 1		
FD-400-PID-002 P&ID sheet 2, control configuration, loop data, SIS C&E				R — TIC-401 loop data	
FD-V1-IN-CE-001 Cause and effect matrix	R — FIC-400 HI alarm row			R — TIC-401 alarm rows	
FD-V1-IN-SRS-001 Safety requirements specification	C — process safety time of SIF-1 (filling rate)	C — PSHH-426 margin to the PSV set	C — PSHH-426 margin to the PSV set		
FD-V1-SA-HZP-001 HAZOP report	C — deviation causes quote the design flow	C — safeguards quote the PSV set	C — safeguards quote the PSV set	C — more-temperature causes	C — HZ-05 hot-oil leak, R-02

Example computed by the tab: design feed rate 16 → 20 m³/h

Action	Document	What changes
revise	FD-400-PFD-001	flags: stream 2 10,208 → 12,760 kg/h, stream 4 1,714 → 2,147, stream 5 8,495 → 10,613 kg/h
revise	FD-400-PID-001	note 1: vapour 1.72 → 2.15 t/h, liquid 8.49 → 10.61 t/h, duty 417 → 525 kW; note 13 FIC-400 HI 18 m³/h below the new flow
revise	FD-V1-PE-HMB-001	maximum-rate case becomes the design case: vapour 611 → 766 Nm³/h, bottoms 14.03 → 17.53 m³/h, E-400 420 → 525 kW (model run, FD-V1-PE-HMB-001)
revise	FD-V1-PE-LIN-001	6 P&ID lines outside the criteria at 20 m³/h: 2"-P-4001-A1A, 2"-P-4002-A1A a, 4"-PV-4021-A1A b, 2"-P-4003-A1A, 2"-HO-4011-B1A, 2"-HO-4012-B1A
revise	FD-V1-ME-EQL-001	E-400 duty 525 kW; FCV-400 opening (model) 73.5 → 82.0 %, LCV-400 63.4 → 71.0 %
revise	FD-V1-ME-DS-001	E-400 rated duty 525 kW, area excess 22.9 % (still ≥ 0); D-400 shell unchanged
revise	FD-V1-IN-INL-001	FT-400 at 80 % of its 0–25 m³/h span; FIC-400 alarms 5 / 20 m³/h sit at the new normal flow → move the HI alarm; FIC-410 17.5 m³/h against 8 / 18

Action	Document	What changes
revise	FD-V1-IN-CE-001	FIC-400 alarm rows (set 20 m³/h) follow the INL
revise	FD-V1-EE-LL-001	P-400 absorbed power 4.04 → 4.77 kW (motor 7.5 kW still enough)
revise	FD-V1-PC-PKG-001	PKG-03: FCV-400 runs choked at ≈ 95 % → larger trim (EI-03); PKG-02 E-400 rating
revise	FD-V1-CM-PCC-001	performance-test guarantees: FI-400 20, FI-420 766 Nm³/h, FI-410 17.53 m³/h, FI-405 14.69 t/h
check	FD-V1-PE-RLB-001	blocked-outlet loads scale with the feed: C2 296 → 370, C3 1,050 → 1,313 kg/h, below the fire case 4,247 kg/h → governing case and orifice J unchanged
check	FD-V1-LA-PLT-001	bottoms flashing at FT-410: 17.5 m³/h is close to the 18 m³/h checked in LA-06 → check the 2" bottoms line
check	FD-V1-IN-SRS-001	SIF-1 filling rate × 1.25: process safety time × 0.80 (LSHH-416 delay 5 s still small)
check	FD-V1-SA-HZP-001	deviation causes quote the design flow; no new deviation

5 · Engineering-query register

Series	Discipline	Raised in tab	Open	Partly answered	Answered	Total
EP	Process	Process engineering	9	1	1	11
EM	Mechanical	Mechanical	4	—	1	5
ER	Relief	Relief & vent	6	1	—	7
EI	Instrumentation	Instrumentation & control	6	—	—	6
EE	Electrical	Electrical	5	—	—	5
EC	Procurement / commissioning	Procurement to commissioning	4	—	—	4
LA	Layout	Layout & isometrics	7	—	—	7
EJ	Project (document control)	Project & deliverables	7	—	1	8
All			48	2	3	53

No.	Status	Discipline	Document	Subject	Query	Why it matters	Proposed answer	Answered by
EP-01	OPEN	Process	FD-V1-PE-LIN-001	Let-down outlet drawn 2"	FD-400-PID-001 draws the line from FCV-400 to D-400 as part of 2"-P-4002-A1A. Downstream of FCV-400 the feed flashes (17 wt% vapour at 102.7 kPa g): in 2" the velocity would be 48.3 m/s and ρv² 63270 Pa at design (60.38 m/s at the maximum rate).	Two-phase flow at that velocity erodes the pipe, vibrates and makes slugs; the drum inlet nozzle is 6" in the Design tab.	Give the let-down outlet its own number at the spec break (FCV-400): 6"-P-4004-A1A (smallest size inside the two-phase criteria at the maximum rate), FCV-400 placed close to D-400, nozzle N1 6".	—
EP-02	OPEN	Process	FD-V1-PE-HMB-001	FCV-400 is in choked flashing service	At design FCV-400 takes ΔP 447.3 kPa while the choked limit is $\Delta P_{max} = F_L^2(P_1 - F_F P_v) = 208.2$ kPa (F_L 0.90 typical). The model uses the square-root law over the whole ΔP.	Choked, the valve needs C_v 10.82 instead of 7.38 (opening 84.7 % instead of 73.5 % on the model's installed C_v 18.17); the trim must take flashing.	Specify FCV-400 for flashing service (angle body, hardened trim, outlet expansion) and size it on the choked ΔP (Instrumentation tab); keep the model as is and teach the difference.	—
EP-03	PARTLY ANSWERED	Process	FD-V1-PE-LIN-001	Bottoms line 2"-P-4003-A1A at the bubble point	D-400 liquid leaves at its bubble point. In 2" it runs at 1.8 m/s (2.25 m/s at the maximum rate) against ≤ 1.0 m/s for bubble-point liquid; the Design tab sizes the outlet nozzle 3".	Friction and acceleration at the nozzle make vapour before LCV-400 (cavitation, unstable level control, FT-410 error).	Make the line 4" from N3 to LCV-400 (XV-418 and FT-410 with it).	**LA-06** (in part): The layout hydraulics confirm flashing at FT-410 in 2" at the high-alarm flow and low level; 4" keeps a positive margin. The size decision is still open.
EP-04	OPEN	Process	FD-V1-PE-HMB-001	LCV-400 outlet flashes	Across LCV-400 the liquid drops from 133.6 to 50 kPa g, below its bubble pressure 102.7 kPa g: 5.3 wt% flashes (stream 5A). LCV-400 is choked (ΔP 83.5 > ΔP_{max} 43.1 kPa). The model keeps the bottoms liquid down to the receiver.	Drawn 2" the outlet line would run at 17.36 m/s, ρv² 18927 Pa.	Number the outlet 4"-P-4005-A1A; specify LCV-400 for flashing; tell the downstream unit that its feed arrives two-phase.	—
EP-05	OPEN	Process	FD-V1-PE-LIN-001	Hot-oil lines carry the full 3-way flow	TCV-401 is a 3-way valve: 2"-HO-4011-B1A and 2"-HO-4012-B1A carry the whole 36 t/h, not only the 11 t/h to E-400. In 2" that is 5.77 m/s and 519.33 kPa/100 m.	The header ΔP of 150 kPa (PDI-406) would be spent in the piping instead of the valve and E-400.	Make the supply and return 4"; the E-400 branches 3"-HO-4013-B1A / 3"-HO-4015-B1A and the bypass 4"-HO-4014-B1A follow the criteria (3").	—
EP-06	OPEN	Process	FD-400-PID-001 note 13	FIC-400 high alarm differs	P&ID note 13 gives FIC-400 alarms 5 / 18 m³/h; the DCS alarm table (ALM) and the Design tab use 5 / 20 m³/h. The maximum-rate case (20 m³/h) sits on the DCS value.	Two documents disagree on an alarm; the operating window with alarms ends at the high alarm.	Confirm 20 m³/h (FIC-400 HI = maximum rate of the HMB) and revise note 13.	—
EP-07	OPEN	Process	FD-V1-PE-LIN-001	Feed system design pressure not stated	The P&ID gives the supply as 600 kPa g at design; the model's P-400 curve shuts off at 650 kPa g. Rule (1) on the shut-off gives 820 kPa g for the feed line, E-400 tubes and FCV-400 (class 150 is enough).	Every item upstream of FCV-400 must hold the pump shut-off; the D-400 side is protected by PSV-400.	State P-400 maximum shut-off (maximum suction + maximum differential) in the design basis; spec break at FCV-400: 820 / 450 kPa g.	—
EP-08	OPEN	Process	FD-V1-PE-HMB-001	Hot-oil header pressures and properties assumed	The model has only the header ΔP (150 kPa) and c_0; supply / return pressures 500 / 350 kPa g, density and viscosity are assumed for the HMB and the line list.	The hot-oil line sizes, E-400 shell design pressure and XV-409 rating depend on them.	Obtain the hot-oil system data sheet (pressures, oil grade and properties) from the utility owner.	—
EP-09	OPEN	Process	FD-V1-IN-INL-001	AI-420 sample at its dew point	The vapour leaves D-400 saturated at 49.2 °C; the sample line to AI-420 at ambient (28 °C) will condense and the GC will read the wrong composition.	Analyser reading drives the spec (AI-420 ≥ 75 wt%).	Heat-trace and insulate the AI-420 sample line to ≥ 60 °C (load HT-420 in EE-LL-001), or reduce the pressure at the tap.	—
EP-10	ANSWERED	Process	FD-V1-PE-LIN-001	2" feed lines above the ΔP criterion	2"-P-4001-A1A and the hot-feed segment of 2"-P-4002-A1A run at 52.05 / 56.85 kPa/100 m at design (80.44 / 88.15 at the maximum rate) against 45. The model's lumped line loss is 10 kPa at design, i.e. about 19 m of 2" pipe.	Line friction eats into the 191 kPa margin between the FCV-400 inlet pressure and the bubble pressure of the hot feed.	Keep 2" only if the routed length stays below ≈ 19 m (Layout tab); otherwise 3".	**FD-V1-LA-PLT-001** (answers): The routed 2" feed (P-4001 + P-4002 to FCV-400) is 51.6 m equivalent as plot.json places XV-408 and 32.4 m with LA-01, above the ≈ 19 m the model's line loss allows: 3" is needed either way.
EP-11	OPEN	Process	FD-V1-PE-LIN-001	Vapour line after PCV-420	Downstream of PCV-420 the vapour (4"-PV-4021-A1A, segment b) expands to the 30 kPa g header: in 4" it runs at 18.8 m/s (23.56 m/s and 14.2 kPa/100 m at the maximum rate).	The header back-pressure plus line loss sets how much ΔP is left for PCV-420 (Operating windows tab: the header limits the vapour valve).	Expand to 6" after PCV-420 (the usual expansion after a gas control valve).	—
EM-01	ANSWERED	Mechanical	FD-V1-ME-DS-001	D-400 elevation datum	The model's K.Hel = 3 m is the static height of the drum bottom (lowest point) above grade, the plot plan gives the bottom tangent at 3 m. The 2:1 head is 0.4 m deep, so one of them is 0.4 m off.	Static head to LCV-400 changes by ≈ 2.4 kPa; NLL elevation for the fire case changes by 0.4 m.	Fix the datum on the GA: bottom tangent 3 m above grade (plot plan), drum bottom 2.6 m; the model keeps 3 m.	**LA-02** (answers): K.Hel is the drum-bottom height above the LCV-400 inlet, not above grade; the plot plan (BT EL 103.000) is kept and the model constant is revised in the next model revision.
EM-02	OPEN	Mechanical	FD-V1-ME-DS-001	"MAWP 450 kPa g" on the P&ID	The vessel calculated with the selected plates (shell 8 mm, heads 10 mm) has a MAWP of 684 kPa g (hot and corroded, top). The P&ID calls 450 kPa g the MAWP; it is the design pressure.	The stamped MAWP and the PSV set must be consistent (UG-134).	Stamp the vessel at the design pressure 450 kPa g @ 150 °C (common practice) and state the calculated MAWP on the datasheet; PSV-400 set 450 kPa g.	—
EM-03	OPEN	Mechanical	FD-V1-ME-DS-001	D-400 nozzles larger than the P&ID lines	Nozzle N1 (feed) is 6" and N3 (liquid) 3" in the Design tab sizing; the P&ID lines are 2". Nozzle N2 (vapour) 4" matches 4"-PV-4021-A1A.	A nozzle sized for the duty with a line two sizes smaller moves the restriction into the line.	Align with EP-01 / EP-03 (let-down outlet and bottoms line sizes) and show the reducers on the next P&ID revision.	—
EM-04	OPEN	Mechanical	FD-V1-ME-DS-001	E-400 thermal design by the vendor	The P&ID gives only the duty (417 kW); the model gives UA 3.89 kW/K clean. The datasheet assumes U 300 W/m² K (Design tab) and a BEU bundle with 16.8 m² (22 % excess at design).	Area, passes and baffles set the tube-side ΔP that the supply curve relies on (40 kPa).	Issue the datasheet for a vendor rating; ask for the clean / fouled U and the ΔP at the maximum rate.	—
EM-05	OPEN	Mechanical	FD-V1-ME-DS-001	No vent / purge connection on D-400	The P&ID shows a drain (HV-415) but no vent, steam-out or utility connection on D-400.	The drum must be gas-freed for entry and purged with N_2 before start-up.	Add a 2" vent with blind on the top head (V1) and a utility connection; route the vent to the flare.	—
ER-01	OPEN	Relief	FD-V1-PE-RLB-001	PSV-400 orifice on HOLD	P&ID note 3 holds the relief cases and size. The governing case is the external fire (4247 kg/h, 656 mm²) → orifice J (830 mm², 2J3); the Design tab also selects J.	The valve cannot be bought without the orifice and the inlet / outlet sizes.	Release the HOLD: PSV-400 2J3 (J, 830 mm²), conventional, set 450 kPa g, inlet 3"-PV-4023-A1A, tail pipe 6"-FL-4041-F1A.	—
ER-02	OPEN	Relief	FD-V1-PE-RLB-001	No thermal relief on the E-400 tube side	XV-408 and FCV-400 both fail closed (I-1, air failure) and can trap the feed in E-400 while hot oil still flows (XV-409 may stay open on a non-SIS closure). Expansion 1.79 m³/h at the start, boiling above 111 °C.	Blocked-in liquid heated by 180 °C oil reaches the tube design pressure in seconds.	Add TSV-400 (¾" × 1", D orifice, set 820 kPa g) on the E-400 tube outlet, discharging to D-400.	—
ER-03	OPEN	Relief	FD-V1-PE-RLB-001	Gas blow-by into the downstream receiver	With the level lost and LCV-400 wide open, 2274 kg/h of drum vapour flows to the 40 kPa g receiver (7837 kg/h at the PSV set, 11211 kg/h with HV-413 open).	The receiver relief belongs to the downstream unit; this unit relies on I-2 (LSLL-417 → XV-418).	Send the blow-by rate to the downstream unit for its relief study; confirm the SIL of I-2 in the SRS.	—
ER-04	OPEN	Relief	FD-V1-PE-RLB-001	Model PSV characteristic	The model's PSV-400 pops at 445 kPa g and passes 0.02 kg/s per kPa above 425 kPa g: 5040 kg/h at 495 kPa g. A J orifice passes 4973 kg/h of blocked-outlet vapour at 495 kPa g and 5377 kg/h in the fire case at 545 kPa g.	The trainee sees PSV-400 lift a little below set and modulate; a real conventional valve pops at set and runs full open.	Keep the model (teaching) and state the difference in the Relief tab; the capacity is of the right order for a J valve.	—

No.	Status	Discipline	Document	Subject	Query	Why it matters	Proposed answer	Answered by
ER-05	OPEN	Relief	FD-V1-PE-RLB-001	Overfill goes to the vapour header	In the model a blocked liquid outlet fills D-400 and carries liquid into the vapour line (AI-420 falls to 35.5 wt%); the pressure does not rise and PSV-400 never sees liquid.	The fuel-gas / vapour header must not receive liquid; I-1 (LSHH-416) is the only barrier.	Confirm a knock-out drum on the vapour header downstream; keep LSHH-416 as the safeguard (HAZOP).	—
ER-06	OPEN	Relief	FD-V1-PE-RLB-001	PCV-430 capacity and restriction orifice	The regulator size is not on the P&ID; with C_v 1.5 (assumed) a wide-open PCV-430 passes 154 kg/h N_2 at the relieving pressure (orifice J holds it easily). The HAZOP (R-15) asks for a restriction orifice.	The N_2 case must stay below the PSV capacity whatever regulator is bought.	State the regulator C_v and size the RO so that the N_2 flow at 700 kPa g ≤ 10 % of the PSV capacity.	—
ER-07	PARTLY ANSWERED	Relief	FD-V1-PE-RLB-001	Flare superimposed back-pressure assumed	The built-up back-pressure (6.2 kPa, 1.4 % of set in 6"; 10.4 % in 4") was calculated with 10 kPa g superimposed in the flare header (assumed) and a 45 m equivalent tail pipe (assumed route).	A conventional valve tolerates ≤ 10 % built-up; a higher header pressure would call for a balanced bellows valve.	Obtain the flare-header back-pressure from the flare study.	**FD-V1-LA-PLT-001** (in part): The routed tail pipe (16.2 m + 3 elbows = 25.4 m equivalent) is shorter than the 45 m assumed; the flare-header back-pressure is still to come from the flare study.
EI-01	OPEN	Instrumentation	FD-V1-IN-INL-001	PT-420 range below the PSV set	PT-420 / PIC-420 span 0–300 kPa g (controller scale). PSV-400 is set at 450 kPa g and relieves at 495 kPa g (overpressure 10 %); between 300 and the set the DCS shows only "over-range". PSHH-426 is a switch: it gives no reading either.	In an overpressure event the operator loses the pressure trend exactly when it matters; the relief study and the incident review need the peak.	Keep PT-420 0–300 kPa g for control resolution and make PSHH-426 a transmitter 0–600 kPa g (trip in the SIS logic, value shown on the DCS), or range PT-420 0–600 kPa g.	—
EI-02	OPEN	Instrumentation	FD-V1-IN-INL-001	PSHH-426 and PT-420 tap the same line	PT-420 and PSHH-426 both connect to 4"-PV-4021-A1A downstream of the vapour nozzle N2. Separate taps, but a plugged nozzle, a closed root valve shared by mistake or liquid in the line affects both. The level SIFs use their own drum nozzles (N7, N8) apart from the LT-400 bridle (N6A/B).	IEC 61511 expects the SIF sensor to be independent of the BPCS initiating cause; pressure rising behind a plugged connection would blind the control and the trip together.	Give PSHH-426 its own nozzle on the D-400 top head (add to the nozzle list with V1, EM-05) and its own root valve, car-sealed open.	—
EI-03	OPEN	Instrumentation	FD-V1-IN-INL-001	FCV-400 rated Cv for choked flow at the maximum rate	Sized on the choked ΔP (EP-02) FCV-400 needs C_v 15.3 at the maximum rate; with the model's rated C_v 18.2 (equal %, R = 30) it would run 94.9 % open, above the 90 % limit. The model, which uses the square-root law over the whole ΔP, shows 82 %.	A valve at its stop at the maximum rate has no control margin left; flashing trims also lose capacity as they wear.	Specify FCV-400 with a flashing-service trim of rated C_v ≥ 32 (opening ≤ 85 % at the maximum rate, 68 % at design); keep the 2" body and expand after the valve (EP-01).	—
EI-04	OPEN	Instrumentation	FD-V1-IN-INL-001	SIS reset and bypass switches not on the P&ID	The SRS asks for a manual reset at the SIS console and for maintenance bypasses under permit with the bypass state shown on the DCS (HAZOP R-28). FD-400-PID-002 shows only HS-400.	Without hard-wired reset and override inputs the reset and bypass are done in software, outside the SIS safety manual; the I/O count misses 4 DI.	Add HS-401 (reset) and HS-416M / HS-426M / HS-417M (key-operated overrides with DCS status) to the P&ID and the INL (rows marked rec.).	—
EI-05	OPEN	Instrumentation	FD-V1-IN-INL-001	TCV-401 valve ΔP versus the model	The model passes hot oil to E-400 in proportion to the TCV-401 opening (linear, 10 kg/s at 100 % with the full 150 kPa header ΔP): C_v 38 for the whole branch. The HMB gives the valve only 20 kPa (H1 → H2, assumed pressures, EP-08), which needs C_v 31.7 for the valve alone at design — the two are not the same valve.	The real 3-way valve and the E-400 shell share the header ΔP; the authority of the valve sets how linear the loop is.	Size TCV-401 with the vendor E-400 shell ΔP and the hot-oil header data; keep the model branch law for teaching and state the difference.	—
EI-06	OPEN	Instrumentation	FD-V1-IN-INL-001	Level transmitter calibrated for one density	LT-400 (remote seals, 3.3 m between taps) is calibrated with the design liquid 605 kg/m³: LRV -29.22, URV -11.55 kPa. A true 50 % reads 51.3 % with the cold start-up liquid (619 kg/m³) and a true 90 % reads 92.2 %.	At start-up the DCS level is high compared with the gauge; operators must know which one to trust (LG-400 is a float gauge: it also depends on density).	State the calibration density on the datasheet; add a density-compensation note to the start-up procedure (compare LT-400 with LG-400 when the drum is cold).	—
EE-01	OPEN	Electrical	FD-V1-EE-LL-001	Analyser shelter not on the plot plan	The load list and the zone plan place the analyser shelter at x 26–30 m, y 1–4 m (assumed); it is not in core/plot.json. There it is 5.8 m from the nearest source: inside the 7.5 m ground-level Zone 2 band.	Shelter location sets the sample-line lengths (tracing loads, analyser lag) and the Ex rating of the shelter equipment.	Layout to fix the shelter on the plot plan; if it stays in Zone 2, pressurise the shelter (or use Ex ec / Ex d equipment) and add a gas detector that trips the shelter power.	—
EE-02	OPEN	Electrical	FD-V1-EE-LL-001	P-400 keeps running against a closed XV-408	I-1 closes XV-408 but P-400 (upstream unit, 7.5 kW) is not stopped: it runs at shut-off (650 kPa g) until someone acts.	A centrifugal pump at shut-off heats its contents; the butane-rich feed can flash in the casing and damage the seal.	Ask the upstream unit for a minimum-flow line on P-400, or add a hard-wired stop of P-400 from the SIS (DO to the upstream MCC) with a run-status DI back.	—
EE-03	OPEN	Electrical	FD-V1-EE-LL-001	SIS UPS autonomy not specified	DCS and SIS take 0.72 kW on the UPS; the documents give no autonomy. 30 min is assumed here (UPS 2 kVA, battery ≈ 0.5 kWh).	After the autonomy the SIS de-energises: the unit trips to the safe state (by design), but the operator loses all indication.	Fix the autonomy (time to bring the unit to a safe, monitored state by hand); consider a dedicated SIS UPS or a separate SIS distribution from the redundant UPS.	—
EE-04	OPEN	Electrical	FD-V1-EE-LL-001	AT-400 sample line also needs tracing	EP-09 proposes tracing for the AT-420 sample line only. AT-400 samples the same saturated vapour (49.2 °C) and condenses at ambient 28 °C in the same way.	Condensate in the N_2 analyser sample reads low N_2; the start-up sweep (AI-400 < 2 %) would be judged on a wrong value.	Trace AT-400 sample line as HT-400 (load list) or share one traced sample line for AT-400 and AT-420.	—
EE-05	OPEN	Electrical	FD-V1-EE-LL-001	No gas detection in the unit	The P&IDs show no flammable-gas detectors. The unit has Zone 1 at the sample point and pits, a Zone 2 band at grade and an analyser shelter inside it.	Detection gives the warning that the zone classification assumes (secondary releases are found and stopped quickly).	Add point gas detectors at the LCV-400 / sample area, at E-400 and in the analyser shelter (trip of the shelter power); count them in the I/O.	—
EC-01	OPEN	Procurement	FD-V1-PC-PKG-001	D-400 sets the critical path	With the typical lead times the critical path runs PKG-01 → SET → PIPE → INL → TEST → INS → MC → PCOM → COMM → PT; mechanical completion is in week 59, handover in week 65. The vessel (requisition week 4, lead 26–35 weeks (typical)) has no float.	Every week lost on the vessel requisition or in the shop moves start-up by a week.	Issue the D-400 requisition first (week 4 at the latest), place the order on the datasheet with nozzles on HOLD for EP-01 / EM-03 if needed, expedite plate and the shop hydrotest.	—
EC-02	OPEN	Commissioning	FD-V1-CM-PCC-001	Test-pack boundaries at the spec breaks	The feed system tests at 1230 kPa g (1.5 × 820) and the hot-oil branch at 1500 kPa g, but D-400 may see no more than its shop test 585 kPa g. FCV-400, LCV-400 and the XVs sit on the breaks.	A test pack that includes D-400 by mistake overpressures the vessel; control valves are not test blinds.	Make test packs with blinds at FCV-400 inlet, LCV-400 outlet and the N_2 regulator; remove or spool out control valves and orifice plates during flushing and testing.	—
EC-03	OPEN	Commissioning	FD-V1-CM-PCC-001	Inerting vent route	The N_2 pressure cycles (2 × 0 → 200 kPa g) must vent air / N_2 somewhere. D-400 has no vent connection (EM-05); through PCV-420 the mixture would go to the fuel-gas (vapour) header.	Air into a fuel-gas header makes a flammable mixture in the header and at the users.	Use the proposed vent V1 to the flare (or a temporary vent to a safe location) for the inerting; keep HV-421 closed until AI-400 shows the drum free of air.	—
EC-04	OPEN	Commissioning	FD-V1-CM-PCC-001	Performance test needs good analysers	The guarantees are AI-420 ≥ 75 wt% (design 76.8, margin 1.8) and AI-410 ≤ 28 wt% (design 26.6, margin 1.4). The margins are of the order of a GC's calibration error.	An uncalibrated or condensing analyser can fail or pass the test on its own error.	Calibrate AT-410 / AT-420 with certified standards before the test, confirm the sample-line tracing (EP-09, EE-04) and take laboratory samples in parallel.	—
LA-01	OPEN	Layout	core/plot.json	XV-408 position on the plot plan	plot.json puts XV-408 at (13, 11), east of E-400 and 2 m west of FCV-400, i.e. on the hot feed between E-400 and FCV-400. FD-400-PID-001 sheet 1 draws XV-408 FC → FT-400 → TI-400 → E-400 → TT-401 → FCV-400 → HV-401 → D-400; the line list carries XV-408 on 2"-P-4001-A1A (cold feed) and the equipment list rates it for 65 °C. To keep plot.json, the cold feed has to run past E-400 to XV-408 and back to the west channel: 2"-P-4001-A1A is routed 25.8 m long.	HAZOP R-02 (thermal relief between XV-408 and FCV-400) and the Consequence-analysis S1 trapped volume both assume the P&ID order; the 3-D model and the field rounds would show XV-408 on the wrong line.	Move XV-408 to (4.6, 7.6), EL 101.000, on 2"-P-4001-A1A near the battery limit, with FT-400 downstream (dashed on the plot plan): the cold feed becomes 10.8 m. Update plot.json and re-run the Consequence-analysis distances for S1 (the release point does not change).	—
LA-02	OPEN	Layout / process	model K.Hel, FD-V1-ME-DS-001 (EM-01)	Static head datum of the model (K.Hel)	In the model K.Hel = 3 m is used only in the LCV-400 pressure balance (head = ρg(K.Hel + h_liquid), h_liquid measured from the drum bottom), so it is the height of the drum bottom above the LCV-400 inlet, not above grade. With the bottom tangent at EL 103.000 (plot.json) the drum bottom is at EL 102.600 and LCV-400 at EL 100.800: 1.80 m.	EM-01 compares K.Hel with the bottom tangent and the drum bottom above grade; neither is the datum the model uses. The difference changes the LCV-400 ΔP at design (part 4).	Answer to EM-01: keep the plot plan (bottom tangent EL 103.000); state in the model notes that K.Hel is the drum-bottom height above LCV-400 and set it to the layout value in the next model revision (the LCV-400 opening at design moves by about 1 %, part 4). Until then the model is 1.2 m optimistic on the bottoms head.	—
LA-03	OPEN	Layout / process	core/plot.json, FD-V1-PE-LIN-001 (EP-01)	Two-phase riser after FCV-400	plot.json puts FCV-400 at grade (EL 101.000); the let-down outlet 6"-P-4004-A1A then rises 5.9 m to N1 (EL 106.900) carrying a flashing two-phase mixture (17 wt% vapour at design).	Typical practice places a let-down valve close to the drum and at about the inlet-nozzle elevation so that the flashing section is short and does not rise (slug flow, vibration, erosion at the elbows).	Keep FCV-400 at grade for operability (FIC-400 and the HV-401/402/403 bypasses are worked from grade) only after a two-phase flow-regime and vibration check of the 6″ riser; otherwise move the FCV-400 station to PL-1 (EL 105.600) with a 2″ hot-liquid riser below it.	—
LA-04	OPEN	Layout / instrumentation	FD-400-PID-001, SRS	LSLL-417 not reachable for proof testing	LSLL-417 sits on N8 at EL 103.450, 3.45 m above grade and below PL-1; the bridle bottom N6B is at EL 103.150. Neither is within 2.0 m of a floor.	The SIS needs LSLL-417 (I-2) proof-tested at the interval of the SRS; work at height without a platform needs a mobile elevating work platform each time.	Add a small access platform on the SE quadrant of the skirt at about EL 102.000 (reached by LD-1), or fit LSLL-417 as a DP / guided-wave transmitter with its electronics at grade.	—
LA-05	OPEN	Layout / safety	core/plot.json	Siting of the control building and the substation	Distances to the nearest hydrocarbon item: control building and substation (part 3, SR-06 to SR-09). The control building lies east, across the prevailing south-west wind, and the substation north-east, downwind of the unit.	Typical spacing tables ask 30–60 m from light-hydrocarbon process equipment to a non-blast-resistant occupied building and about 30 m to ignition sources, with both upwind. The Consequence-analysis tab finds the control building inside the LFL cloud of S1, S2 and S5.	Building siting study (blast, gas ingress, fire) before the layout is frozen; move the substation upwind (south-west) or provide gas detection at its air intakes; the control building as in the Consequence-analysis findings.	—
LA-06	OPEN	Layout / process	FD-V1-PE-LIN-001 (EP-03)	Bottoms line flashes at FT-410 at high flow and low level	With 2"-P-4003-A1A the static head between the drum liquid and FT-410 does not cover the entrance, friction and orifice losses at the FIC-410 high-alarm flow and a low drum level: the vena contracta of FT-410 drops below the bubble pressure (part 4).	Flashing in the orifice gives a false (low, noisy) flow reading to FIC-410 / LIC-400 (Rev B cascade) and cavitation damage.	Make the bottoms line 4″ from N3 to LCV-400 (EP-03), keep FT-410 at the lowest point of the run, size the orifice for a low ΔP (≤ 10 kPa at full scale) or use a magnetic / Coriolis meter.	—
LA-07	OPEN	Layout / instrumentation	FD-400-PID-001	Field items without a tag number	The control-valve station block valves (P&ID note 5: locked open) and the local gauges the Field tab reads (PG-420 on the drum top, the hot-oil supply gauge) have no tag on the P&ID; the layout calls them BV-FCV400-U/D, BV-LCV400-U/D, BV-PCV420-U/D, PG-420 and PG-406.	Field rounds, isolation lists and the 3-D model need a unique tag for every valve and gauge the operator touches or reads.	Number them on the next P&ID revision (for example HV-404…HV-409 for the station blocks, PI-420 / PI-406 for the gauges) and update layout.json.	—

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EJ-01	OPEN	Project	FD-V1-PM-DOC-001	Two numbering schemes and two status words	The PFD and the two P&ID sheets carry the older numbers FD-400-PFD-001 / FD-400-PID-001 / -002; every later document uses FD-V1-(discipline)-(type)-001. The register in epc/epc.json gives the PFD and the P&ID sheets the status "Issued for training", every other document "Issued for teaching".	A register exists so that one number and one status identify one document; two schemes make cross-references (and the transmittals) error-prone.	Keep the drawing numbers (they are printed on the drawings and quoted by the HAZOP, C&E and SRS) and list them in this register under their own numbers; use "Issued for teaching" for all documents (this register already does) and correct the title blocks at Rev 1.	—
EJ-02	OPEN	Project	FD-V1-IN-SRS-001	The SRS is not a controlled file	FD-V1-IN-SRS-001 exists only as the SRS tab of the page; there is no issued PDF with a revision table, unlike the C&E (FD-V1-IN-CE-001) and the HAZOP report (FD-V1-SA-HZP-001).	The SRS is the document the SIS vendor designs to and the FAT is tested against (FD-V1-CM-PCC-001); it must be issued, revised and transmitted like the others.	Issue FD-V1-IN-SRS-001 as a PDF (Rev 0, Issued for teaching) from the SRS tab data, with the recommended SIF additions marked as HOLD.	—
EJ-03	OPEN	Project	Consequence analysis (study)	Consequence screening used line sizes assumed before the line list	The Consequence-analysis tab assumed a 3" vapour line (77.9 mm) and a 2" bottom nozzle (52.5 mm) because no line list existed. The line list now has 4"-PV-4021-A1A (4", 102.3 mm) and the datasheet nozzle N3 3" (77.9 mm). Scenario S5 (full-bore bottom-nozzle leak) scales with d^2: $(77.9 / 52.5)^2 = 2.20\times$ the release rate. S3 uses the 25 mm RAST hole and is not affected.	S5 already screens at severity 5; a larger release lengthens the LFL distance and the pool-fire radius used in the siting findings (LA-05) and the LOPA.	Re-run S5 with the 3" N3 nozzle (and S3 full bore with 4" if a full-bore case is wanted) once EP-03 fixes the bottoms line; update the HAZID/LOPA severity notes.	—
EJ-04	OPEN	Project	FD-400-PID-001 / -002	P&ID Rev 1 collects 18 open queries	18 queries propose a change to the P&ID sheets: EP-01, EP-03, EP-04, EP-05, EP-06, EP-11, EM-02, EM-03, EM-05, ER-01, ER-02, ER-06, EI-02, EI-04, EE-05, LA-01, LA-07, EJ-07. Until they are answered, the documents built from the P&ID (line list, instrument list, isometrics, packages) carry the Rev 0 values with a pointer to the query.	A P&ID revised one query at a time is re-issued many times and each re-issue ripples through every document built from it (part 4); late P&ID changes are the most common cause of rework in detailed engineering.	Cloud the affected items on the Rev 0 sheets as HOLD; answer the queries in one review; issue FD-400-PID-001 / -002 Rev 1 once, then revise the line list, instrument list and isometrics in one pass.	—
EJ-05	OPEN	Project	FD-V1-LA-PLT-001	Isometrics drawn at sizes that an open query changes	The isometrics use the P&ID line sizes. 2"-P-4001-A1A (ISO-1) needs 3" at the maximum rate (EP-10); 2"-P-4002-A1A segment a (ISO-2) needs 3" at the maximum rate (EP-10); 4"-PV-4021-A1A segment b (ISO-3) needs 6" at the maximum rate (EP-11); 2"-P-4003-A1A (ISO-4) needs 4" at the maximum rate (EP-03).	An isometric released for fabrication at the wrong size is scrap; spools are prefabricated weeks before the line is erected.	Mark ISO-1, ISO-2, ISO-3, ISO-4 "HOLD — not for fabrication" until EP-10, EP-11, EP-03 are answered; the other isometrics may be released.	—
EJ-06	ANSWERED	Project	FD-V1-PE-RLB-001	Two fire relief loads for PSV-400	The Design tab gives a fire load of 3943 kg/h (λ 299 kJ/kg, 604 mm²); the relief load basis gives 4247 kg/h (λ 277.55 kJ/kg from the model at the relieving pressure, 656 mm²). Both select orifice J.	Two documents quoting two loads for the same valve confuse the vendor and the reviewer.	The relief load basis FD-V1-PE-RLB-001 governs (it is the controlled document); the Design tab is a preliminary hand calculation and keeps its value with a note.	**FD-V1-PE-RLB-001** (answers): Answered by document hierarchy: the relief load basis governs; same orifice letter, so no document downstream changes.
EJ-07	OPEN	Project	FD-400-PID-001, FD-V1-IN-CE-001	D-400 can drain back through FCV-400 after a trip	I-1 and the ESD close XV-408, XV-409 and XV-418 but not FCV-400; with the feed stopped FIC-400 drives FCV-400 open on zero flow. A leak at the FCV-400 inlet (Consequence-analysis S1) is then fed back from D-400 (2,517 kg of liquid at 50 % level, Consequence-analysis design state) through the open valve. The only check valve (HAZOP R-03) sits upstream of E-400.	The trip isolates the feed side but not the drum inventory; the trapped volume assumed for S1 does not hold.	Add a check valve on the let-down line at N1 (6"-P-4004-A1A) or let I-1 de-energise the FCV-400 solenoid (fail closed); add the effect to the C&E and the SRS.	—
EJ-08	OPEN	Project	FD-V1-PE-RLB-001, FD-V1-LA-PLT-001	Fire case assumes drainage that the layout does not show	The relief load basis takes the API 521 fire heat input with adequate drainage and fire fighting: 327 kW, 4247 kg/h, orifice J. Without drainage the same wetted area gives 537 kW, 6970 kg/h, about 1076 mm² → orifice K. The Consequence-analysis S5 bottom-nozzle leak forms a 356 m² pool with 37.5 kW/m² out to 19.9 m: it engulfs D-400, and the plot plan shows no kerb or sloped paving leading spills away from the drum.	The smaller heat input is only valid if a pool cannot form under the vessel; otherwise PSV-400 is one orifice size too small for the fire case.	Show kerbed, sloped paving draining away from D-400 to a remote sump on the plot plan (Safety-layers item KERB), or size PSV-400 for the no-drainage heat input; record the decision in the relief load basis.	—

6 · Cross-document checks

Computed by the Project & deliverables tab from the register data, the model, the DCS alarm table, the C&E / SRS data and the PFD data. A ✗ is a finding, recorded by the query in the last column.

Number checked	Documents	First value	Second value	Result	Query
Stream 2 (feed) mass flow: HMB = PFD flag = model	HMB · PFD · model	HMB 10,208 · PFD 10,208 kg/h	model 10,208 kg/h	✓	
Stream 4 (vapour) mass flow: HMB = PFD flag = model	HMB · PFD · model	HMB 1,714 · PFD 1,714 kg/h	model 1,714 kg/h	✓	
Stream 5 (liquid) mass flow: HMB = PFD flag = model	HMB · PFD · model	HMB 8,495 · PFD 8,491 kg/h	model 8,496 kg/h	✓	
Hot feed: HMB stream 3 = PFD flag = TIC-401 set point	HMB · PFD · model	HMB 74.0 °C, 550 kPa g · PFD 74 °C, 550 kPa g	K.TICsp 74 °C; FCV-400 inlet 550 kPa g	✓	
Hot oil to E-400: HMB H2 = PFD flag = FI-405	HMB · PFD	HMB 10,980 kg/h · PFD 10,980	FI-405 10,980 kg/h	✓	
E-400 duty: HMB = model	HMB · model	420.2 kW	420.2 kW	✓	
P&ID general note 1 = HMB (vapour, liquid, duty) within 1 %	P&ID · HMB	1.72 t/h, 8.49 t/h, 417 kW	1.71 t/h, 8.49 t/h, 420 kW	✓	
FIC-400 high alarm: P&ID note 13 = DCS alarm table	P&ID · ALM · INL	note 13: 5 / 18 m³/h	ALM 5 / 20 m³/h	✗	EP-06
D-400 design pressure ≥ PSV-400 set (equipment list vs model)	EQL · RLB · model	450 kPa g	K.Ppsv 450 kPa g	✓	
Drum-side items (LCV-400, PCV-420, XV-418, PSV-400 inlet) design ≥ PSV set	EQL	LCV-400 450, PCV-420 450, XV-418 450, PSV-400 450	≥ 450 kPa g	✓	
Feed-side items (E-400 tubes, FCV-400, XV-408) design ≥ P-400 shut-off	EQL · model	E-400 820, FCV-400 820, XV-408 820	≥ 650 kPa g (K.Psup0)	✓	EP-07
D-400 calculated MAWP ≥ stamped MAWP ≥ PSV set (UG-134)	DS · RLB	684 ≥ 450	≥ 450 kPa g	✓	EM-02
Instrument list alarm limits = DCS alarm table (ALM)	INL · ALM	11 of 11 tags equal	all equal	✓	
Instrument list trip settings = model (K.Lshh, K.Pshh, K.LsII)	INL · model	LSHH-416 90, PSHH-426 250, LSLL-417 5	90 / 250 / 5	✓	
PT-420 range covers the PSV-400 set	INL · RLB	0–300 kPa g	set 450 kPa g	✗	EI-01
Relief: governing case area ≤ selected orifice; orifice capacity ≥ governing load	RLB	C6 fire: 656 mm², 4,247 kg/h	J 830 mm², 5,377 kg/h	✓	ER-01
Relief: the governing case is the largest credible vapour load to PSV-400	RLB	C2 296, C6 4,247	governing C6	✓	
Relief: fire load recomputed in this page = relief load basis	RLB · model	4,247 kg/h, 655 mm²	4,247 kg/h, 656 mm²	✓	
PSV-400 tail pipe and inlet: line list = relief basis	LIN · RLB	6"-FL-4041-F1A, 3"-PV-4023-A1A	tail 6" (built-up), inlet 3"	✓	
Isometric lines: size and design P/T = line list	PLT · LIN	10 of 10 lines on ISO-1…5	all equal	✓	
Isometrics drawn at sizes no open query changes	PLT · LIN	2"-P-4001-A1A (ISO-1), 2"-P-4002-A1A a (ISO-2), 4"-PV-4021-A1A b (ISO-3), 2"-P-4003-A1A (ISO-4)	3" EP-10, 3" EP-10, 6" EP-11, 4" EP-03	✗	EJ-05
D-400 nozzle size = size of the line connected to it	DS · LIN	N1 6", N2 4", N3 3", N4 2", N5 1"	N3 3" on 2"-P-4003-A1A	✗	EM-03, EP-03
C&E trip rows = SRS SIFs (every SIF sensor has a C&E row, every trip row a SIF)	CE · SRS	4 trip rows: HS-400, LSHH-416, PSHH-426, LSLL-417	3 SIFs: SIF-1 LSHH-416, SIF-2 PSHH-426, SIF-3 LSLL-417	✓	
C&E trip settings = model	CE · model	LSHH-416 90, PSHH-426 250, LSLL-417 5	90 / 250 / 5	✓	
C&E alarm rows = DCS alarm table	CE · ALM	21 alarm rows	all equal	✓	
P-400 absorbed power: load list = HMB	LL · HMB	4.04 kW	4.04 kW	✓	
DCS cabinet load = installed DCS I/O × 10 W + 150 W (load list vs instrument list)	LL · INL	380 W	23 I/O → 380 W	✓	
Every item of the equipment list is in a purchase package	EQL · PKG	11 of 11	all	✓	
D-400 hydrotest: checklist = datasheet; leak test = 0.9 × PSV set	PCC · DS · RLB	585 / 405 kPa g	585 / 405 kPa g	✓	
Consequence analysis line sizes = line list / datasheet	cq · LIN · DS	vapour 77.9 mm, bottom nozzle 52.5 mm (assumed)	4"-PV-4021-A1A 102.3 mm, N3 3" 77.9 mm	✗	EJ-03

25 ✓, 5 ✗ of 30 checks.

Register integrity

Check	Value	Against	Result	Query
Every register file is in a tab folder or published with the page	14 of 14	all found	✓	
Every tab named in the register exists in this page	17 of 17	eproj, eproc, erel, emech, einst, eelec, ecomm, lay, pfd, pid, cem, srs, hazop	✓	
Document numbers unique	17 numbers	no duplicates	✓	
Document numbers follow FD-V1-(discipline)-(type)-001	14 of 17	FD-400-PFD-001, FD-400-PID-001, FD-400-PID-002	✗	EJ-01
One status word in the register data	14 of 17 "Issued for teaching"	FD-400-PFD-001: Issued for training, FD-400-PID-001: Issued for training, FD-400-PID-002: Issued for training	✗	EJ-01
Query numbers unique	53 queries	no duplicates	✓	
Every query has a number, a discipline and a document	53 of 53	all	✓	
Every document number cited by a query is in the register	13 of 13	all	✓	
Every query number cited inside another query exists	25 of 25	all	✓	
Every "answered by" is a query or a register document	5 of 5	all resolve	✓	
Every query links to a tab present in this page	9 of 10	cq	—	
Every register document is a controlled file	14 of 17	FD-400-PID-001 (drawing in the tab), FD-400-PID-002 (drawing in the tab), FD-V1-IN-SRS-001 (tab only)	✗	EJ-02
Every dependency joins two register documents	41 of 41	all	✓	
The dependency tree has no loop (each document can be issued after its inputs)	no loop	17 documents, 41 dependencies	✓	

7 · Findings of the study tabs that point to a design query

Source tab	Finding	Design point	Documents	Query (status)
Consequence analysis	**Finding 1** — HZ-01 and HZ-17 were ranked S4 in the HAZID; the screening gives N ≈ 4 and 2.7 (flashing hot feed / drum liquid, clouds of 150–250 m over the unit and the control building), i.e. S5.	HAZID / HAZOP severity to revise (safety action, no design change)	FD-V1-SA-HZP-001	safety action in the study tab
Consequence analysis	**Finding 2** — The control building (4 people, not blast-resistant) lies inside the LFL cloud of S1, S2 and S5; S6 shows the HVAC drawing a flammable mixture in within minutes.	building siting, gas detection at the air intake with HVAC trip	FD-V1-LA-PLT-001, FD-V1-EE-LL-001	LA-05 (open), EE-05 (open)
Consequence analysis	**Finding 3** — I-1 / ESD do not close FCV-400 and FIC-400 opens it on zero flow: the drum liquid can drain back to a leak at the FCV-400 inlet.	check valve at N1 or FCV-400 closed by I-1	FD-400-PID-001, FD-V1-IN-CE-001	EJ-07 (open)
Consequence analysis	**Finding 4** — HAZOP R-02 puts E-400 between XV-408 and FCV-400; the plot plan puts XV-408 between E-400 and FCV-400.	move XV-408 on the plot plan	FD-V1-LA-PLT-001	LA-01 (open)
Consequence analysis	**Finding 5** — A D-400 bottom-nozzle leak is not isolable and its pool fire engulfs D-400.	kerbed paving draining away from D-400, or a no-drainage fire case for PSV-400; S5 to re-run with the 3" nozzle	FD-V1-PE-RLB-001, FD-V1-LA-PLT-001	EJ-08 (open), EJ-03 (open)
Consequence analysis	**Finding 6** — The flare (≈ 99 m from FCV-400) lies inside the S1 and S5 LFL distances: a delayed ignition source.	siting of the flare and the occupied buildings	FD-V1-LA-PLT-001	LA-05 (open)
Consequence analysis	**Finding 7** — The LOPA has no scenario for line leaks screened at S5.	LOPA scope (safety action)	FD-V1-IN-SRS-001	safety action in the study tab
Incident cases	**Grangemouth 1987** — Gas broke through a level valve into a low-pressure separator whose relief was sized for fire only.	gas blow-by through LCV-400 / HV-413 to the downstream receiver	FD-V1-PE-RLB-001	ER-03 (open)
Incident cases	**Milford Haven 1994** — A valve shown open was shut; the unit overfilled into the flare system.	overfill to the vapour header, knock-out drum at the users	FD-V1-PE-RLB-001	ER-05 (open)
Incident cases	**Texas City 2005** — The tower was overfilled at start-up with a level transmitter reading wrong.	LT-400 calibration density at start-up	FD-V1-IN-INL-001	EI-06 (open)
Incident cases	**Buncefield 2005** — A stuck gauge and an independent high-level switch that was never tested.	proof testing of the level switches needs access	FD-V1-LA-PLT-001, FD-V1-IN-SRS-001	LA-04 (open)
Incident cases	**Valero McKee 2007** — A propane release that could not be isolated remotely; no gas detection.	no gas detection in the unit	FD-V1-EE-LL-001	EE-05 (open)
Incident cases	**Williams Geismar 2013** — A blocked-in exchanger heated by its hot side ruptured.	no thermal relief on the E-400 tube side	FD-V1-PE-RLB-001	ER-02 (open)
Incident cases	**BP-Husky Toledo 2022** — A fuel-gas mix drum overfilled; liquid reached the users.	carry-over to the fuel-gas users	FD-V1-PE-RLB-001	ER-05 (open)